Introduction

Before we proceed to further statistical concepts and calculations, we need to have a very good knowledge on some more new mathematical concepts, rules and formulae to understand in a better way the theory and problems in statistics. Hence we introduce some algebraic methods and elementary calculus in this chapter.

7.1 Fundamental Principles of counting

We use two fundamental principles of counting in solving the problems. They are addition rule on counting and multiplication rule on counting.

Fundamental Principle of addition on counting:

If an operation can be performed in m ways and if another operation can be performed in n ways and only one operation can be done at a time, then either of the two operations can be done at a time can be performed in $m + n$ ways.

Example 7.1

In a box there are 5 red balls and 6 green balls. A person wants to select either a red ball or a green ball. In how many ways can the person make this selection?.

Solution:

Selection of a red ball from 5 balls in 5 ways.

Selection of a green ball from 6 balls in 6 ways.

By the fundamental Principle of addition, selection of a red ball or a green ball in (5+6) =11 ways.

Fundamental Principle of multiplication on counting:

If an operation can be performed in m ways and if another operation in n ways independent of the first, then the number of ways of performing both the operations simultaneously in m × n ways.

Example 7.2

A person has to travel from a place A to C through B. From A to B there are two routes and from B to C three routes. In how many ways can he travel from A to C?.

Solution:

